2. Modern biotechnology

There are two main features of this technology, that differentiated it from the conventional technology they are i) ability to change the genetic material for getting new products with specific requirement through recombinant DNA technology ii) ownership of the newly developed technology and its social impact. Today, biotechnology is a billion dollar business around the world, where in pharmaceutical companies, breweries, agro industries and other biotechnology based industries apply biotechnological tools for their product improvement.

Modern biotechnology embraces all methods of genetic modification by recombinant DNA and cell fusion technology. The major focus of biotechnology are

- **Fermentation** for production of acids, enzymes, alcohols, antibiotics, fine chemicals, vitamins and toxins
- **Biomass for** bulk production of single cell protein , alcohol, and biofuel
- **Enzymes** as biosensors, in processing industry
- **Biofuels** for production of hydrogen, alcohol, methane
- **Microbial inoculants** as biofertiliser, and nitrogen fixers
- **Plant and animal cell culture** for production of secondary metabolites, monoclonal antibodies
- **Recombinant DNA technology** for production of fine chemicals, enzymes, vaccines, growth hormones, antibiotics, and interferon
- **Process engineering** – tools of biotechnology is used for effluent treatment, water recycling.

This unit will reveal the various aspects of modern biotechnology, its products and applications.

4.3. Methods of Biotechnology

4.3.1 Fermentation

The word fermentation is derived from the Latin verb 'fervere' which means ' to boil'. Fermentation refers to the metabolic process in which organic molecules (normally glucose) are converted into acids, gases, or alcohol in the absence of oxygen or any electron transport chain. The study of fermentation, its practical uses is called zymology and originated in 1856, when French chemist Louis Pasteur demonstrated that fermentation was caused by yeast. Fermentation occurs in certain types of bacteria and fungi that require an oxygen-free environment to live. The processes of fermentation are valuable to the food and beverage industries, with the conversion of sugar into ethanol to produce alcoholic beverages, the release of CO_2 by yeast used in the leavening of bread, and with the production of organic acids to preserve and flavor vegetables and dairy products.

Bioreactor (Fermentor)

Bioreactor (Fermentor) is a vessel or a container that is designed in such a way that it can provide an optimum environment in which microorganisms or their enzymes interact with a substrate to produce the required product. In the bioreactor aeration, agitation, temperature and pH are controlled. Fermentation involves two process namely upstream and downstream process.

i. Upstream process

All the process before starting of the fermenter such as sterilization of the fermenter, preparation and sterilization of culture medium and growth of the suitable inoculum are called upstream process.

ii. Downstream process

All the process after the fermentation process is known as the downstream process. This process includes distillation, centrifuging,

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